Analysis

Baseline

The baseline bag-of-words model had an accuracy of 0.6325. This is close to the expected 65% target, so it worked reasonably well as a starting point.

Stopword Removal

Stopword removal did not improve performance. Accuracy dropped from 0.6325 to 0.6195.

This may be because short tweets often use small words that still matter for sentiment, such as negation words. Removing too many words may have removed useful information.

Unigrams + Bigrams

The unigram + bigram model did improve performance. Accuracy increased from 0.6325 to 0.6503, which is an increase of about 0.0178.

This likely helped because bigrams can capture short phrases, not just single words. Sentiment often depends on phrases like “not good” or “very happy.”

Stopwords + Bigrams

The stopwords + bigrams model was the best model. Accuracy increased from 0.6325 to 0.6533, which is an improvement of about 0.0208.

This model probably worked best because it used phrase information from bigrams while also removing some less useful words.

spaCy Embeddings

The spaCy embeddings model did not improve performance. Accuracy dropped from 0.6325 to 0.5807.

This may be because averaging word vectors loses details that are important in tweets, such as slang, short phrases, hashtags, and negation. A better embedding model or more tweet-specific preprocessing could improve this.